A-A-54316
14 January 1991

COMMERCIAL ITEM DESCRIPTION

DRILL, TWIST, BONE, Round Shaft

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers corrosion-resistant steel twist drills suitable for use in orthopedic surgery

Salient Characteristics:

Style, design, and dimensions. The drills shall be of one-piece construction. The proximal end of the drills shall be round and non-notched for use in Jacobs Chuck type hand and power drills. The flutes of the drills shall be uniformly cut, shaped, and sharpened to cut at the tip as well as the sides of the operative end and provide automatic chip clearance The drills shall conform to Table I for dimensions for the applicable size

Table I Drill dimensions

Size	Diameter + 0, -.005 (-0, -0 1)	Flute Length + 125 (+ 3.2)	Helix Angle + 3 °	Overall Length + 0, -.125 (+0, -3.2)
1	.094 inch (2.1 mm)	.975 inch (24.8 mm)	21 °	5 inch (127 0 mm)
2	.063 inch (1.6 mm)	.975 inch (24.8 mm)	19 °	5 inch (127.0 mm)
3	078 inch (2 0 mm)	.975 inch (24 8 mm)	21	5 inch (127 0 mm)
4	.109 inch (2 8 mm)	1.250 inch (31.8 mm)	23	5 inch (127.0 mm)
5	125 inch (3 2 mm)	1 250 inch (31 8 mm)	24	5 inch (127.0 mm)

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Table I Drill dimensions (continued)

Size	Diameter + 0,- 005 (-0,-0 1)	Flute Length + .125 (+ 3.2)	Helix Angle + 3°	Overall Length + 0, -.125 (+0,-3 2)
6	.141 inch (3 6 mm)	1.250 inch (31 8 mm)	25 °	5 inch (127.0 mm)
7	156 inch (4.0 mm)	1 250 inch (31.8 mm)	25 °	5 inch (127 0 mm)
8	.188 inch (4.8 mm)	1.375 inch (34.8 mm)	25 °	5 inch (127.0 mm)
9	.172 inch (4.4 mm)	1.375 inch (34.8 mm)	25 °	5 inch (127.0 mm)

Chisel edge angle shall be 13° ± 2°.

Material. The drills shall be fabricated from type 420C or 440B stainless steel in accordance with ASTM Designation F899.

Hardness The drills shall have a Rockwell Hardness C scale (HRC) of 53 to 58 when tested in accordance with ASTM Designation E18 or E92 whichever is applicable Conversion of hardness values shall be in accordance with ASTM Designation E140.

Finish. The finish of the drills shall be mirror, except the operative ends shall have a fine honed finish with sharp cutting edges. As an alternate, twist drills shall be electropolished to a surface finish that is less reflective.

Corrosion resistance. The drills shall show no evidence of corrosion-resistance when subjected to the boil test and copper sulfate test specified in ASTM Designation F1089.

Runout. The total indicated runout shall not exceed .001 inch per inch when tested as specified herein.

Sterilization. The drills shall be capable of withstanding repeated sterilization via steam, ethylene oxide, or long term 10 hour soaking in cold sterilization solution. The drills shall show no evidence of corrosion when autoclaved as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Workmanship. The drills shall be free from burrs or other defects which detract from their appearance or impair their serviceability

Unit Package (PG) One package containing six drills of one size only, as specified, constitutes one unit

QUALITY ASSURANCE PROVISIONS

Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level II, AQL 2.5 except for dimensional examination, the inspection level shall be S-2, AQL 2 5. The unit of product shall be one drill

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-2, AQL of 1 0 The unit of product for test purposes shall be one drill

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Runout. The drill shall be placed and held in a V block or precision chuck, and rotated slowly. A dial indicator sensitive to .0002 inch and equipped with a knife-edge foot, a shadowgraph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the drill, and so positioned that the foot contacts the drill shank nearest the head flutes. The maximum and minimum readings obtained during one complete 360 degrees rotation of the drill shall be measured and recorded to the nearest .0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Autoclaving. The test drills shall be cleaned and placed in an autoclave. The drills shall then be autoclaved at 250 degrees F ± 5 degree F and 27 ± 1 pounds per square inch pressure for a minimum of 5 minutes. After the drills are removed from the autoclave, the entire surface area of the drill shall be examined for evidence of corrosion.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Regulatory requirements

Federal Food, Drug, and Cosmetic Act If the product covered by this document has been determined by the U S Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug Cosmetic Act, as amended, and regulations promulgated thereunder In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal State, and local statutes, ordinances, and regulations

Recovered materials The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc , involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and /or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION	Size
6515-01-142-8219	DRILL, TWIST, BONE, Round Shaft, 3/32 inch diameter, 5 inch length, 6s	1
6515-01-142-8220	DRILL, TWIST, BONE, Round Shaft, 1/16 inch diameter, 5 inch length, 6s	2
6515-01-142-8221	DRILL, TWIST, BONE, Round Shaft, 5/64 inch diameter, 5 inch length, 6s	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

6515-01-142-8222	DRILL, TWIST, BONE, Round Shaft, 7/64 inch diameter, 5 inch length, 6s	4
6515-01-142-8223	DRILL, TWIST, BONE, Round Shaft, 1/8 inch diameter, 5 inch length, 6s	5
6515-01-142-8224	DRILL, TWIST, BONE, Round Shaft, 9/64 inch diameter, 5 inch length, 6s	6
6515-01-142-8225	DRILL, TWIST, BONE, Round Shaft, 5/32 inch diameter, 5 inch length, 6s	7
6515-01-142-8226	DRILL, TWIST, BONE, Round Shaft, 3/16 inch diameter, 5 inch length, 6s	8
6515-01-143-2050	DRILL TWIST, BONE, Round Shaft, 11/64 inch diameter, 5 inch length, 6s	9

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order)

MILITARY INTERESTS

Custodians.

Army - MD
Navy - MS
Air Force - 03

Preparing Activity

DoD-MB

Agent.

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4679

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
Defense Personnel Support Center
ATTN: DPSC-MSTE
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-54316	DRILL, TWIST, BONE, Round Shaft

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.